

Learning, Interaction, and Impact

Final Week



Agenda:

- Presentation on
- Demo on Perplexity

What about using this for the shared interest thing

5 Core Ideas

- **Perception**
 - Process of extracting meaning from sensory signals. Making computers “see” and “hear” well enough for practical use
- **Representation and reasoning**
 - AI systems build simple models of the world using data. These models help them make decisions, but they do not think like humans.
- **Learning**
 - AI systems get better by finding patterns in large amounts of data. This process helps them make predictions.
- **Natural interaction**
 - AI systems need many types of knowledge to communicate naturally with people (language structure, ambiguity, context & memory limits, applications, emotion recognition, commonsense gaps)
- **Societal impact**
 - Positive impacts, unequal effects, bias and fairness, transparency, trust, economic effects, ethical design

- **Perception**
 - “This is how AI takes in what you give it.”
 - “For example, if you paste a long message and ask ‘summarize this,’ the AI is first reading and breaking down that text.”
 - “Or if you speak into a voice tool and it turns it into text, that’s perception.”
 - “It’s just about taking input and making it usable.”
- **Representation and reasoning**
 - “This is how AI decides what to say back.”
 - “For example, if you ask ‘write a professional email,’ it figures out what ‘professional’ should sound like and builds a response.”
 - “Or if you say ‘compare these two options,’ it organizes the info and lays it out clearly.”
 - “This is the decision-making part of AI.”

- Learning

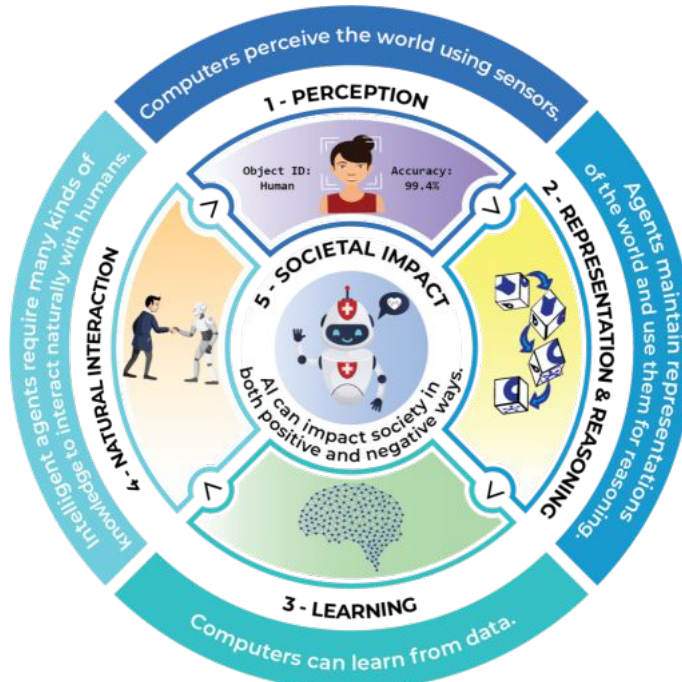
- “This is how the AI was trained before you even use it.”
- “For example, ChatGPT didn’t just appear — it was trained on tons of examples of writing, questions, and answers.”
- “That’s why it can do things like write essays or explain concepts.”
- “The learning already happened beforehand.”

- Natural interaction

- “This is why using AI feels so easy.”
- “You don’t need to learn commands — you just talk normally.”
- “For example, you can say ‘make this shorter’ or ‘explain this simply,’ and it understands what you mean.”
- “It’s designed to feel like a conversation.”

- Societal impact

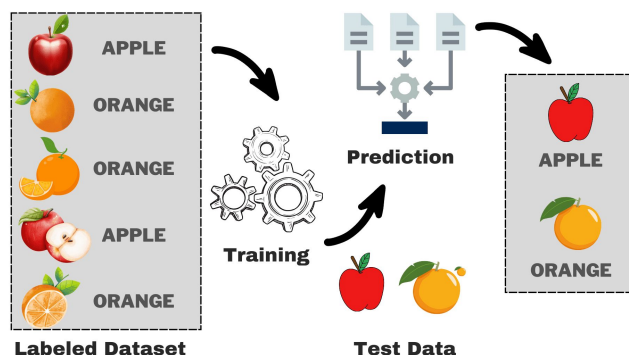
- “This is about how AI affects real life.”
- “For example, students might use it to help with homework, or people use it at work to save time.”
- “But it also raises questions — like what happens if people rely on it too much, or if it gives wrong information.”
- “So it’s not just about the tech — it’s about how we use it.”



Learning

Learning From Examples

- Learning means finding patterns in examples
- The examples shape what the system can do
- Training and using are not the same thing



For this part, we are talking about how GenAI learns. It does not learn exactly like a person. A person can learn by asking questions, watching someone, practicing, making mistakes, and connecting something new to a real life experience.

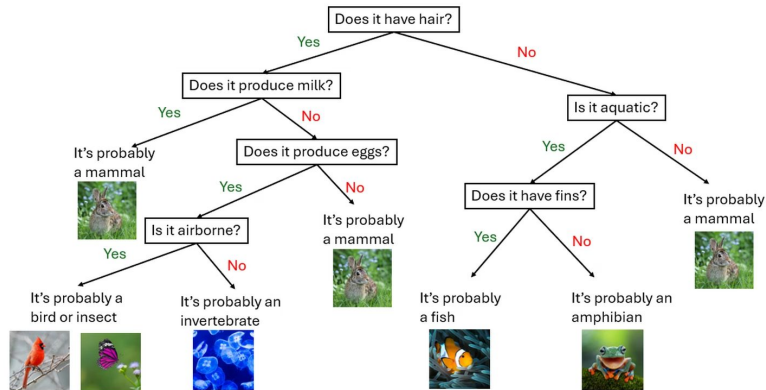
A computer learns in a different way. It looks at many examples and finds patterns. For example, a spam filter learns from emails that were marked as “spam” and “not spam.” Over time, it starts to notice patterns, like suspicious links, urgent wording, strange sender addresses, or repeated phrases.

Training is the stage where the tool learns from examples. Using the tool is different. After training, it

can look at a new email, message, image, or prompt and make a prediction based on what it learned before. So when we use GenAI, we are usually using a model that has already been trained, not teaching it from the beginning each time.

What Changes During Learning?

- Training adjusts the inside of the model
- A decision tree adds questions
- A neural network adjusts weights



When a model learns, something inside it changes. It is not just memorizing one answer. It is adjusting the way it connects inputs to outputs.

A decision tree is easier to picture. Imagine a guessing game where the tool is trying to identify an animal. It might ask: Does it have fur? Does it fly? Does it live in water? Each question helps narrow down the answer. If the answer is wrong, the tree may need a better question.

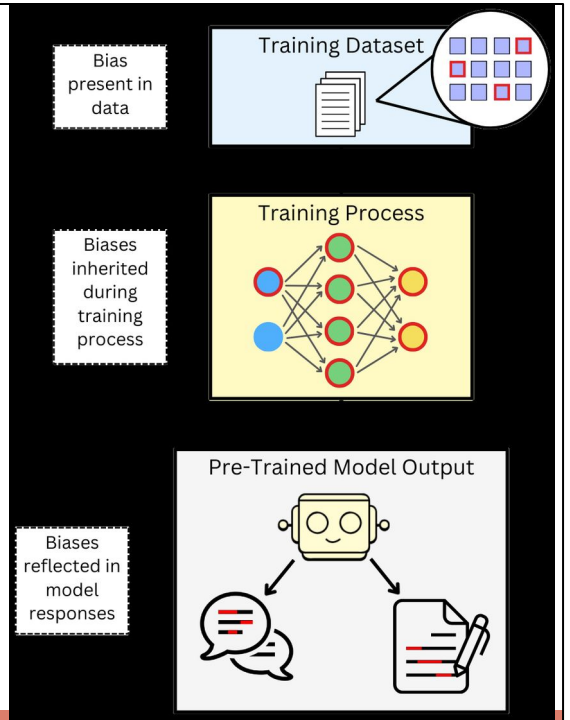
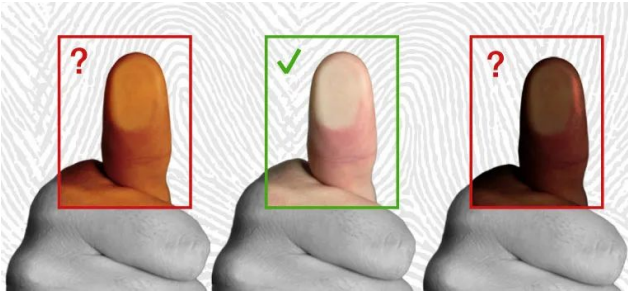
A neural network is harder to see, but the idea is still about adjustment. Instead of clear yes-or-no

questions, it adjusts many small number values inside the model. These small changes help it get closer to the right answer over many examples.

This matters because training data affects what gets adjusted. If the examples change, the model may learn a different pattern. That is why the data behind the tool is so important.

When Learning Goes Wrong

- Bad examples can teach the wrong pattern
- Missing examples can leave people out
- Bias can come from the training data



Learning from examples is powerful, but it can also cause problems. The tool may learn a pattern that works in the training examples but fails in real life.

For example, imagine we train a system to recognize healthy and unhealthy foods. If all the healthy examples are green foods like broccoli, spinach, and peas, and the unhealthy examples are donuts, cake, and candy, the system might accidentally learn that “green means healthy.” Then it may think green candy is healthy too. The problem is not that the tool is trying to be unfair or careless. It learned from examples that were too limited.

The same thing can happen with people, language,

accents, culture, or communities. If the training examples mostly represent one group, the tool may work better for that group and worse for others. That is why GenAI can sometimes give a polished answer that still misses the real situation.

For our setting, this matters with translation, church language, and cultural examples. A tool may know common Christian wording but miss details that are specific to the Eritrean Orthodox community. So when the topic is personal, cultural, religious, medical, legal, or financial, we should slow down and check the answer.

Natural Interaction

Talking to Technology

- Natural interaction means communicating in human ways
- Chatbots can answer, summarize, translate, and write
- Human language still creates problems



Natural interaction is about communicating with technology in a way that feels normal to people. Instead of using exact commands or clicking through many menus, we can type or speak in everyday language.

That is why tools like ChatGPT, Siri, Alexa, Google Translate, and voice assistants feel different from older computer tools. You can ask a question, request a summary, translate a sentence, draft a message, or ask for an explanation.

But natural conversation is not just about words.

People use tone, culture, shared experiences, jokes, indirect meaning, and body language. We also leave things unsaid because we assume the other person understands the situation. GenAI can be helpful with language, but it does not always have the full context that a person would have.

So when a prompt is unclear, the tool fills in missing details using patterns. Sometimes that works. Other times, it gives an answer that sounds smooth but does not fit what we really meant.

Why Language Is Hard

- The same sentence can have more than one meaning
- Context helps people understand what is meant
- Tools may choose the most common meaning, not the right one



Language is difficult because the same words can mean different things depending on the situation.

For example, think about this sentence: "John saw the man with the telescope." That could mean John used a telescope to see the man. Or it could mean the man had the telescope. The sentence does not clearly tell us which meaning is correct.

People usually use context to figure it out. If we were talking about birdwatching, maybe John had the telescope. If we were talking about a man carrying equipment, maybe the man had it. The words alone

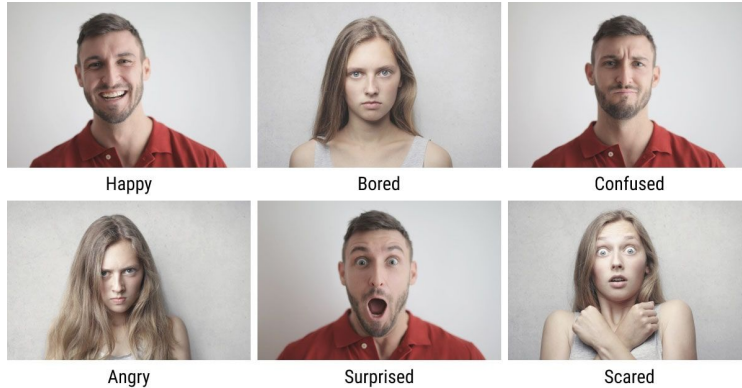
are not always enough.

This is one reason translation can be tricky. A word may have one meaning in everyday speech and another meaning in church, family, or cultural settings. If we ask for a translation without context, GenAI may choose the most common meaning instead of the most appropriate one.

For better results, we should give the situation. Instead of only saying, “Translate this,” we can say, “Translate this for an Eritrean Orthodox church announcement in a respectful tone.” That gives the tool a better chance of choosing the right wording.

Emotion Is Not the Same as Understanding

- Tools can detect emotional tone
- They do not actually feel emotions
- Human support still matters



Some tools can recognize emotional tone. They might look at the words in a message and decide whether it sounds happy, angry, sad, worried, or frustrated. Some systems can also use facial expressions, tone of voice, or body language as clues.

That can be useful. For example, a customer service chatbot might notice that someone sounds upset and respond more carefully. A tutoring tool might notice frustration and suggest slowing down, reviewing a step, or taking a short break.

But recognizing emotion is not the same as feeling emotion. If a chatbot says, “I’m sorry you’re going through that,” it may sound caring, but it is not

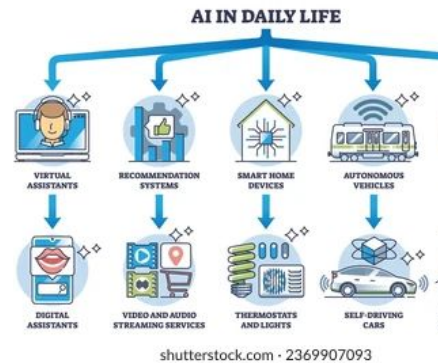
experiencing sadness, love, concern, or empathy like a person does.

That does not mean the tool is useless. It can help us write a kind message, organize our thoughts, or practice what to say. But for serious emotional, spiritual, family, health, or personal situations, real human support still matters. The tool can assist, but it should not replace people.

Societal Impact

AI Is Already Part of Daily Life

- It affects how we search, work, learn, and communicate
- It can make services faster and easier
- It can also change expectations and habits



Societal impact means AI is not just something happening on a computer screen. It affects daily life. It changes how we search for information, how we write, how companies provide customer service, how schools think about homework, how hospitals use data, and how people find directions, jobs, and services.

A helpful example is healthcare. Johns Hopkins developed an AI system to help detect sepsis earlier. Sepsis is dangerous because it can get worse quickly, and early warning can give doctors more time to respond. In a large hospital study, the system helped catch sepsis earlier than traditional methods and was connected to lower death rates for patients.

That is the positive side. AI can help people notice patterns in large amounts of information faster than a person could do by hand. But even with a helpful example like this, people still have to ask careful questions. Who checks the result? What happens if the tool is wrong? Does it work equally well for different groups of people?

So the point is not just that AI is useful. The point is that AI is already entering important parts of life, and we need to understand both the help and the risk.

Fairness, Privacy, and Trust

- A system can work well for some people and poorly for others
- Private information needs protection
- People should know who is responsible



- Trust is a big part of societal impact. If a tool is helping decide something important, we should ask: Is it fair? Is it using the right information? Can someone explain the decision? Who is responsible if it causes harm?
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- A real healthcare example shows why this matters. Researchers found racial bias in a widely used health care risk algorithm. The tool used health care spending as a shortcut for health need. But spending is not the same as sickness. If one group has less access to care, less money may be spent on them even when they

- are very sick. That meant the tool underestimated the needs of some Black patients. Correcting the issue would have increased the share of Black patients receiving extra help from 17.7% to 46.5%.
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- Another example is Air Canada's chatbot. A customer asked about a bereavement fare, and the chatbot gave wrong information. The airline was still held responsible because the chatbot was part of the company's website. That matters because organizations cannot simply say, "The bot made the mistake." If they use the tool, they still have responsibility for how it affects people.
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- This is why privacy and trust matter. We should be careful about putting sensitive information into tools, especially medical, financial, legal, or personal details. And when an answer affects real people, there needs to be human responsibility behind it.
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AI Can Help, But It Can Also Harm

- AI can support social good
- It can also spread harm faster
- Responsible use means looking at the full impact



- AI can be used for social good. It can help with public health, environmental protection, emergency response, accessibility, education, and many other areas.
- One positive example is wildfire detection. Some fire agencies use AI with cameras and sensors to look for smoke or unusual activity. The goal is to detect fires earlier, especially in remote areas where a person may not see the fire right away. California's ALERTCalifornia network uses AI with more than 1,000 cameras to support wildfire detection and emergency response.
- Another positive example is environmental

- monitoring. The United Nations Environment Programme uses satellite data and machine learning to detect methane releases from oil and gas installations. Methane is a powerful greenhouse gas, so detecting leaks can help governments and companies respond faster.
- But the same kind of technology can also be misused. In New Hampshire, an AI-generated robocall imitated President Biden's voice and told voters not to vote in the primary. The FCC later issued a \$6 million fine connected to those robocalls.
- So the lesson is not "AI is good" or "AI is bad." The better question is: who is using it, for what purpose, with what data, and who might be helped or harmed?
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Data Centers: Where AI Actually Runs

- AI depends on physical infrastructure
- Northern Virginia is a major data center region
- Energy, water, land, and noise affect communities



AI may feel like it lives on our phones or laptops, but it actually runs through physical buildings called data centers. These are large facilities filled with servers that process and store data.

They support everyday tools like email, cloud storage, streaming, apps, websites, and AI. Northern Virginia is a major example close to us. Virginia historically holds the title for the largest concentration of data center capacity in the world. Northern Virginia "Data Center Alley"

There are benefits, like jobs, tax revenue, construction, and technology investment. But there are also concerns about electricity use, water, land,

noise, and pollution. That is why Prince George's County paused data center approvals after pushback over the old Landover Mall site.

Memphis is another example. Residents and environmental groups raised concerns about xAI's data center using gas turbines and increasing pollution in nearby communities.

So data centers show that AI is not just digital. It has a real footprint in communities.

<https://www.datacentermap.com/content/nova/>

The Landover Mall Data Center Debate

- Proposed for the old Landover Mall site
- County paused approvals after community pushback
- Debate includes jobs, tax revenue, water, power, pollution, and neighborhood impact



A local example is the fight over the proposed data center at the old Landover Mall site in Prince George's County, just off the Beltway.

Residents want the empty land developed, but many do not want a data center there. Supporters point to tax revenue, investment, and the fact that data centers support everyday tools like phones, email, streaming, cloud storage, and AI.

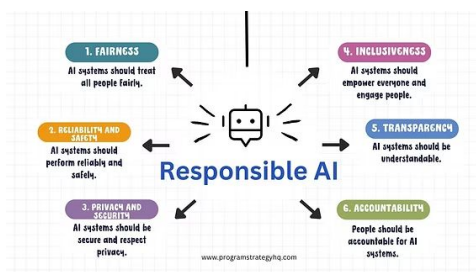
Residents and advocates raised concerns about water pressure, sediment in the water, diesel generator pollution, energy demand, and long-term neighborhood impact. Because of the pushback, county leaders decided to hold off on approvals for

now.

This shows that technology can be useful, but we still have to ask where it goes, who it helps, who it harms, and whether the community gets a real say.

What Should We Remember?

- Learning depends on data
- Natural conversation still needs context
- AI has real social and community impacts
- Keep using AI carefully, creatively, and responsibly



Since this is our last class, the main thing I want you to leave with is that AI is useful, but it is not magic.

It learns from data, so the examples matter. If the data is limited or missing certain people, languages, or communities, the answers can be limited too.

It can talk in a natural way, but it still needs context. The more clearly we explain what we need, who it is for, and what situation we are in, the better the response usually is.

And finally, AI affects the real world. It can help with learning, writing, translation, healthcare, and daily tasks, but it also raises questions about fairness, privacy, jobs, energy use, and community impact.

So after this workshop, the goal is not to know everything about AI. The goal is to feel more comfortable using it, asking better questions, checking important answers, and thinking carefully about how it affects people.

Questions?

Passcode

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Attendance



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